The converse on the boundary divisors at full q, and why the A-type divisors have none: the residue of a structure constant is a difference across the collision pairing

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Abstract

Rick assigned me, in writing in UIDs 827 and 828, *"the boundary/A-type (S/A) divisors at full q"* — the one family his Theorem D^{∞} explicitly does not settle. The first finding is that **the assignment's own grouping is wrong**: the two halves have different answers.

For the A-type divisors $\{w_i = w_j\}$ there is **no converse to be had**. Theorem B of `2026-10-18-atype` holds for *every* unit Ψ of the A-type GKM ring, so no renormalisation whatever produces an A-type pole: the right-hand side of any "pole $\iff$ condition" statement is identically false. That is a result, not a failure, and it is recorded as such.

For the boundary divisors $S^{\varepsilon}_m = \{w_m = \varepsilon\}$, $\varepsilon^h = 1$, there *is* a converse, and this paper proves it. The mechanism is a single identity. Writing $\gamma^{\nu}_{\mathbf{M}}$ for the residue at S of the $(\nu, \mathbf{M})$ entry of Φ^{-1}, the boundary collapse law forces

$$\boxed{\gamma^{\nu}_{\mathbf{R}\cup\{m\}} = -\,\gamma^{\nu}_{\mathbf{R}}} \qquad \text{for every } \mathbf{R} \subseteq [N] \smallsetminus \{m\},$$

so the residue functional is a *difference across the collision pairing*. Feeding $f = \Psi_\chi \phi_\lambda \phi_\mu$ through it collapses 2^N terms to a sum over

$$E^S(\lambda, \mu, \nu; m) := \{\mathbf{R} \subseteq [N] \smallsetminus \{m\} \ : \ \mathbf{R} \cup \{m\} \le \nu,\ \mathbf{R} \ge \lambda,\ \mathbf{R} \ge \mu\},$$

with prefactor $1 - \chi(\varepsilon)$ and every term individually nonzero (Theorem 17). Hence: no pole if $\chi(\varepsilon) = 1$ or $E^S = \varnothing$; a pole, necessarily simple, only if $E^S \neq \varnothing$; and — with the quantifier where it belongs —

$$E^S(\lambda, \mu, \nu; m) \neq \varnothing \iff \textit{some admissible } \chi \textit{ with } \chi(\varepsilon) \neq 1 \textit{ makes } \hat{c}^{\nu}_{\lambda\mu} \textit{ singular at } S^{\varepsilon}_m,$$

which is Theorem 19, proved by a Vandermonde independence of the squarefree monomials $\prod_{r \in \mathbf{R}} \chi(w_r)$. For a *fixed* χ the one remaining step is that $|E^S| \ge 2$ nonzero terms do not

cancel; that is the exact analogue of what Rick needed Theorem W and D^∞ for, it is stated as the single gap, and $|E^S| \geq 2$ occurs in 38% ($N = 3$) to 87% ($N = 6$) of the firing triples, so the census is not drawn from inside its own hypothesis class.

Two by-products. **(i)** Motegi's order is the *tail-counting* order $\mathbf{B} \leq \mathbf{A} \iff \#\{c \in \mathbf{B} : c \geq k\} \leq \#\{c \in \mathbf{A} : c \geq k\}\ \forall k$ (Lemma 5); transitivity, $\mathbf{A} \smallsetminus \{m\} \leq \mathbf{A}$, and **complementation as an order anti-automorphism** all drop out, and the last is what makes Motegi's inverse formula's support match the down-set. **(ii)** $\Delta^X_{\mathbf{A}}$ of `eq:allcard-Delta-factor` (`2610.15344` L4257–4262) *is* my diagonal entry $\widetilde{A}_{\mathbf{A},\mathbf{A}}$, so his all-cardinality biorthogonality gives Φ^{-1} in my own notation; I verified it first-hand against my symmetrisation engine, 336/336.

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1 Scope determination, done first

The brief forbade assuming the target was open, and required a paragraph on *what exactly is open* before any attempt. Here it is.

family	root	untwisted $c^\nu_{\lambda\mu}$	twisted $\hat{c}^\nu_{\lambda\mu}$
pair $\{w_a w_b = 1\}$	$e_a + e_b$	poles $\iff E \neq \varnothing$ (Rick, D^∞)	none iff (COB)
boundary $\{w_m^h = 1\}$	$h\, e_m$	**none**	**one-directional only**
A-type $\{w_i = w_j\}$	$e_i - e_j$	none	**none, for every unit Ψ**
(Q) $\{1 - q \cdot \text{mono}\}$	—	none	none

Row 3 is Theorem B of `2026-10-18-atype-divisors-and-the-gkm-lattice.tex`; row 4 is Theorem `thm:Q` of `2026-10-18-c2`; the untwisted entry of row 2 is `thm:SRC` + `prop:short` of `2026-10-17-c2`; the twisted entry of row 2 is `thm:C`(2) there.

Two consequences, and they point in opposite directions.

The A half is closed and admits no converse. Theorem B of `2026-10-18-atype` is stated for an arbitrary unit Ψ of the A-type GKM ring $\mathcal{G}_{ij}$, not merely for a point-indexed product; and `lem:unit` there puts Ψ_χ in $\mathcal{G}_{ij}^\times$ for every ij-admissible χ. So *no* renormalisation produces an A-type pole. A statement of the form "pole at $\{w_i = w_j\} \iff E^A \neq \varnothing$" therefore holds with E^A identically empty, which is to say it has no content. §8 records this with its mechanism.

The S half is open in exactly one place. `thm:C`(2) says: regular if $\chi(\varepsilon) = 1$; and if $\chi(\varepsilon) \neq 1$, regular *only when* $(\phi_\lambda\phi_\mu)(\mathbf{M}^-)|_S = 0$ for all $\mathbf{M} \ni m$. Measured against Rick's pair-divisor theorem it is missing two things: it is a per-(λ,μ) statement ("*every* $\hat{c}^\nu$ is regular"), not a per-ν one; and the non-vanishing direction is not made combinatorial. So the target of this paper is **the per-ν converse on the boundary family at full q.**

Remark 1 (the arm-3 detector fired, and where)**.** The brief's not-well-posed arm came with a detector: *before measuring, check each candidate divisor is a root hyperplane of the relevant type.* It fires, and it fires on the S half rather than the A half. In type B the boundary root is e_m and the locus is $\{w_m = 1\}$: the second component $\{w_m = -1\}$ of type C's $\{w_m^2 = 1\}$ is **not** a root hyperplane of B_N, and the measurement in `2026-10-19-typeB-gauge-theorem.tex` (§"Beyond the root system") is that the collapse law itself fails there (0/9 at $r \geq 1$, against 6/6 at $r = 0$ where there is no head slot). So every statement below is quantified over $\varepsilon^h = 1$ — two components in type C, one in type B — and the type-B half of the $\varepsilon = -1$ story does not exist rather than being unproved.

2 Setup

Fix $N \geq 1$ and $X \in \{B, C\}$, $h = h_X$ (2 for C, 1 for B). Let $F := \mathbb{C}(q, w_1, \ldots, w_N)$, $\mathcal{P} := 2^{[N]}$ the T-fixed points of $LG(N, 2N)$, and $\mathcal{A} := F^{\mathcal{P}}$ the GKM algebra under pointwise operations. For $\mathbf{I} \in \mathcal{P}$ let $\phi_{\mathbf{I}}(\mathbf{M}) := \widetilde{A}^X_{\mathbf{M},\mathbf{I}}$, Motegi's normalised type-X weight function at $u = w_{\mathbf{M}}$ (`2610.15344`; the engine transcribes `Motherfunction` L473–487 and the normalisers L1057–1062, L1361–1367). Write $\Phi := (\phi_{\mathbf{I}}(\mathbf{M}))_{\mathbf{M},\mathbf{I}}$.

We use three facts from Motegi, all deep-read:

(M3) Triangularity (Thm 2.6(i), L603–620; order defined L582–589). $\widetilde{A}_{\mathbf{M},\mathbf{I}} = 0$ unless $\mathbf{M} \geq \mathbf{I}$, where $\mathbf{M} \geq \mathbf{I}$ means $|\mathbf{M}| \geq |\mathbf{I}|$ and $m_{s-r+j} \geq i_j$ for $j = 1, \ldots, r$ ($s = |\mathbf{M}|$, $r = |\mathbf{I}|$, both sets listed increasingly).

(M4) The diagonal (`normalizationWC` L1078, `normalizationWB` L1386). $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ is a product over the positive roots, with factors $\prod_{j \leq k}(1 - w_{i_j}^{-1} w_{i_k}^{-1})$ over $\mathbf{I}$ — note $j \leq k$, so the boundary term $(1 - w_i^{-h})$ occurs — together with A-type factors and q-factors $(1 - q \cdot \text{monomial})$.

(M5) $\widetilde{A}_{\mathbf{M},\mathbf{I}} \in R := \mathbb{C}(q)[w^{\pm}]$ (Prop. L3302–3309 type C, L3722–3729 type B).

By (M3)+(M4), $\{\phi_{\mathbf{I}}\}$ is an F-basis of $\mathcal{A}$; the structure constants are $\phi_\lambda\phi_\mu = \sum_\nu c^\nu_{\lambda\mu}\phi_\nu$, and $\ell_\nu \in \mathcal{A}^*$ denotes the ν-th coordinate functional, so that $\ell_\nu(f) = \sum_{\mathbf{M}} (\Phi^{-1})_{\nu\mathbf{M}} f(\mathbf{M})$.

Fix $m \in [N]$ and ε with $\varepsilon^h = 1$. Let $S = S^\varepsilon_m := \{w_m = \varepsilon\}$, with local ring $\mathcal{O} \subset F$, uniformiser $t := w_m - \varepsilon$ and residue field k; $\bar{x}$ denotes the image of $x \in \mathcal{O}$ in k. For $\mathbf{M} \ni m$ write $\mathbf{M}^- := \mathbf{M} \smallsetminus \{m\}$. Note $k \cong \mathbb{C}(q)(w_j : j \neq m)$.

(BC) Boundary collapse. For every $\mathbf{I}$ and every $\mathbf{M} \ni m$, $\widetilde{A}^X_{\mathbf{M},\mathbf{I}}\big|_S = \widetilde{A}^X_{\mathbf{M}^-,\mathbf{I}}\big|_S$: the boundary collision multiplier is exactly 1.

(BD) Boundary diagonal. $\operatorname{ord}_S \widetilde{A}^X_{\mathbf{I},\mathbf{I}} = [\, m \in \mathbf{I}\,]$ for every $\mathbf{I}$, at a point of S generic in the sense of Remark 4.

(BC) in type C is `thm:SRC` of `2026-10-17-c2` (proved there in four steps, with a plant at $\varepsilon = 2$ firing 24/24); in type B it is the gauge theorem of `2026-10-19-typeB` at $\alpha = e_m$, where $\rho^\alpha = \psi_B(w_m)^{-1}$ and $\psi_B(v) = (1-qv)/(1-q)$ equals 1 on $\{w_m = 1\}$. (BD) in type C is the count inside `prop:short` of `2026-10-17-c2`; in type B it is **measured here and not proved** (34/34, §9), and every statement using it in type B is graded accordingly in §10.

Definition 2 (admissible renormalisation)**.** $\chi \in \mathbb{C}(q)(v)^\times$ is **S-admissible** if $\chi(w_j)$ is regular and non-vanishing at the generic point of S for every $j \in [N]$; then $\Psi_\chi(\mathbf{M}) := \prod_{m' \in \mathbf{M}} \chi(w_{m'})$ lies in $(\mathcal{O}^\times)^{\mathcal{P}}$. Put $\kappa := \chi(\varepsilon)$, $\hat\phi_{\mathbf{I}} := \Psi_\chi \phi_{\mathbf{I}}$, and let $\hat c^\nu_{\lambda\mu}$ be the structure constants of $\{\hat\phi_{\mathbf{I}}\}$.

Lemma 3 (the twisted constants are coordinates of one function)**.** $\hat c^\nu_{\lambda\mu} = \ell_\nu(f)$ *with* $f := \Psi_\chi\, \phi_\lambda \phi_\mu$.

Proof. $\Psi := \Psi_\chi$ is invertible in $\mathcal{A}$, so $\hat\phi_\lambda \hat\phi_\mu = \sum \hat c^\nu \hat\phi_\nu$ is equivalent to $\Psi^2 \phi_\lambda \phi_\mu = \Psi \sum \hat c^\nu \phi_\nu$, i.e. to $\Psi \phi_\lambda \phi_\mu = \sum \hat c^\nu \phi_\nu$. (This is the first step of `thm:B` of `2026-10-17-c2`.) □

Remark 4 (what "generic" has to mean, and the control that taught me)**.** (BD) is *not* a statement about q alone. The positive-root product $\widetilde{A}_{\mathbf{I},\mathbf{I}}$ contains factors $(1 - q w_m w_k)$ and $(1 - q w_k / w_m)$; at $w_m = \varepsilon$ these are $(1 - q\varepsilon w_k)$ and $(1 - q\varepsilon^{-1} w_k)$, which vanish on the *hypersurfaces* $w_k = \pm q^{-1}$ in the remaining variables. My first $N = 3$ census drew the random point $w_3 = 7/3$ with $q = 3/7$, landed on exactly that hypersurface, and reported $\operatorname{ord}_S \det \Phi = 8$ against the predicted 4 — a confident, legible, wholly false number, caught only because I had required the count as a control. So "generic" here means: outside the finitely many hypersurfaces on which a root factor other than $(1 - w_m^{-h})$ vanishes identically on S. All computations below sample until (BD) holds for every $\mathbf{I}$, verified against an *independent* closed form for the diagonal (the root product `diag_rootproduct`, not the symmetrisation sum), and assert it.

3 Motegi's order is the tail-counting order

Lemma 5. *For* $\mathbf{A}, \mathbf{B} \subseteq [N]$ *write* $\mathbf{B}_{\geq k} := \#\{c \in \mathbf{B} : c \geq k\}$. *Then*

$$\mathbf{B} \leq \mathbf{A} \iff \mathbf{B}_{\geq k} \leq \mathbf{A}_{\geq k} \quad \text{for every } k \in [N].$$

Proof. Write $\mathbf{A} = \{a_1 < \cdots < a_s\}$, $\mathbf{B} = \{b_1 < \cdots < b_r\}$; the relation $\mathbf{B} \leq \mathbf{A}$ is "$r \leq s$ and $a_{s-r+j} \geq b_j$ for $j = 1, \dots, r$".

($\Leftarrow$) Taking $k = 1$ gives $r \leq s$. Fix $j \leq r$ and put $k = b_j$. The elements $b_j, \dots, b_r$ are all $\geq k$, so $\mathbf{B}_{\geq k} \geq r - j + 1$, hence $\mathbf{A}_{\geq k} \geq r - j + 1$, i.e. at least $r - j + 1$ elements of $\mathbf{A}$ are $\geq b_j$; the $(r-j+1)$-th largest of these is $a_{s-(r-j+1)+1} = a_{s-r+j}$, so $a_{s-r+j} \geq b_j$.

($\Rightarrow$) Fix k and put $i := \mathbf{B}_{\geq k}$. If $i = 0$ there is nothing to prove. Otherwise $b_{r-i+1} \geq k$, and taking $j = r - i + 1$ in the hypothesis gives $a_{s-i+1} = a_{s-r+j} \geq b_{r-i+1} \geq k$; hence $a_{s-i+1}, \dots, a_s$ are all $\geq k$ and $\mathbf{A}_{\geq k} \geq i$. □

Corollary 6. *1.* $\leq$ *is a partial order on* $\mathcal{P}$.

2. $\mathbf{A} \setminus \{m\} \leq \mathbf{A}$ *for every* $\mathbf{A} \ni m$; *hence every down-set* $\downarrow\nu := \{\mathbf{M} : \mathbf{M} \leq \nu\}$ *is closed under* $\mathbf{M} \mapsto \mathbf{M}^-$.

3. (complementation is an order anti-automorphism) $\mathbf{B} \leq \mathbf{A} \iff \mathbf{A}^{\mathrm{c}} \leq \mathbf{B}^{\mathrm{c}}$.

Proof. (1) The displayed criterion is a conjunction of inequalities between integers, so it is reflexive and transitive; and $\mathbf{B} \le \mathbf{A} \le \mathbf{B}$ forces $\mathbf{B}_{\ge k} = \mathbf{A}_{\ge k}$ for all k, whence $\mathbf{A} = \mathbf{B}$ (the counting function determines the set: $k \in \mathbf{A} \iff \mathbf{A}_{\ge k} > \mathbf{A}_{\ge k+1}$). (2) Deleting an element decreases every $\mathbf{A}_{\ge k}$ by 0 or 1. (3) $(\mathbf{A}^{\mathrm{c}})_{\ge k} = (N-k+1) - \mathbf{A}_{\ge k}$, so $\mathbf{A}^{\mathrm{c}}_{\ge k} \le \mathbf{B}^{\mathrm{c}}_{\ge k}$ for all k is literally $\mathbf{B}_{\ge k} \le \mathbf{A}_{\ge k}$ for all k. □

Remark 7. Corollary 6(2) is the closure my review of 2026-10-19 flagged as "the first thing to check" when I proposed to transport Rick's down-set argument to the S and A divisors. It holds. Corollary 6(3) is the fact that makes the support of Motegi's inverse formula (Lemma 12) agree with the chain-expansion support of Lemma 10: the formula gives $\widetilde{A}_{\mathbf{M}^{\mathrm{c}},\nu^{\mathrm{c}}} = 0$ unless $\nu^{\mathrm{c}} \le \mathbf{M}^{\mathrm{c}}$, which is $\mathbf{M} \le \nu$ *only because* complementation reverses the order. Two mechanisms, same support, and the anti-automorphism is the bridge. Verified exhaustively: 21844/21844 pairs $(\mathbf{M}, \nu)$, all $N \le 7$.

4 The residue functional, and the identity that drives everything

Throughout this section χ is S-admissible and $\nu \in \mathcal{P}$ is fixed.

Proposition 8 (boundary lattice lemma; `prop:short`)**.** *Assume* (BC) *and* (BD). *Then*

$$\mathcal{L} := \mathrm{span}_{\mathcal{O}}\{\phi_{\mathbf{I}} : \mathbf{I} \in \mathcal{P}\} \;=\; W := \{f \in \mathcal{O}^{\mathcal{P}} \;:\; f(\mathbf{M}) \equiv f(\mathbf{M}^{-}) \pmod{t} \;\; \forall\, \mathbf{M} \ni m\},$$

and $t\,\mathcal{O}^{\mathcal{P}} \subseteq \mathcal{L}$.

Proof. $\mathcal{L} \subseteq W$: the $\phi_{\mathbf{I}}$ lie in $\mathcal{O}^{\mathcal{P}}$ by (M5) and satisfy the congruence by (BC); W is an $\mathcal{O}$-module. Colength of W: the defect map $D : \mathcal{O}^{\mathcal{P}} \to k^{\{\mathbf{M} \ni m\}}$, $D_{\mathbf{M}}(f) := \overline{f(\mathbf{M})} - \overline{f(\mathbf{M}^{-})}$, is $\mathcal{O}$-linear with kernel W, and is surjective — for $\mathbf{M}_0 \ni m$ the indicator of $\mathbf{M}_0$ has $D_{\mathbf{M}} = \delta_{\mathbf{M},\mathbf{M}_0}$, since $\mathbf{M}^{-} = \mathbf{M}_0$ is impossible for $m \in \mathbf{M}_0$. So $\mathrm{length}(\mathcal{O}^{\mathcal{P}}/W) = 2^{N-1}$. Colength of $\mathcal{L}$: over the DVR $\mathcal{O}$ it is $\mathrm{ord}_S \det \Phi = \sum_{\mathbf{I}} \mathrm{ord}_S\, \widetilde{A}_{\mathbf{I},\mathbf{I}}$ (by (M3) and Corollary 6(1): a permutation σ with $\sigma(\mathbf{M}) \le \mathbf{M}$ for all $\mathbf{M}$ is the identity, so $\det \Phi = \prod_{\mathbf{I}} \widetilde{A}_{\mathbf{I},\mathbf{I}}$), which is $\#\{\mathbf{I} \ni m\} = 2^{N-1}$ by (BD). Nested submodules of equal finite colength coincide. Finally $t\,\mathcal{O}^{\mathcal{P}} \subseteq W$ because $\overline{tg(\mathbf{M})} = 0 = \overline{tg(\mathbf{M}^{-})}$. □

Lemma 9 (the residue functional)**.** $t\,(\Phi^{-1})_{\nu\mathbf{M}} \in \mathcal{O}$ *for all* $\mathbf{M}$. *Hence, setting* $\gamma^{\nu}_{\mathbf{M}} := \overline{t\,(\Phi^{-1})_{\nu\mathbf{M}}} \in k$,

$$\mathrm{Res}_{\nu}(f) := \overline{t\,\ell_{\nu}(f)} = \sum_{\mathbf{M} \in \mathcal{P}} \gamma^{\nu}_{\mathbf{M}}\, \overline{f(\mathbf{M})} \qquad \text{for every } f \in \mathcal{O}^{\mathcal{P}},$$

and $\ell_{\nu}(f) \in \mathcal{O} \iff \mathrm{Res}_{\nu}(f) = 0$.

Proof. By Proposition 8, $t\,\mathcal{O}^{\mathcal{P}} \subseteq \mathcal{L}$, so for every $g \in \mathcal{O}^{\mathcal{P}}$ all coordinates $\ell_{\mathbf{I}}(tg)$ lie in $\mathcal{O}$; taking g the indicator of $\mathbf{M}$ gives $t(\Phi^{-1})_{\mathbf{IM}} \in \mathcal{O}$. Consequently $\mathrm{ord}_S\, \ell_{\nu}(f) \ge -1$ for $f \in \mathcal{O}^{\mathcal{P}}$, so $t\ell_{\nu}(f) \in \mathcal{O}$ and its class is $\sum_{\mathbf{M}} \overline{t(\Phi^{-1})_{\nu\mathbf{M}}}\, \overline{f(\mathbf{M})}$. The last claim is the statement that a pole of order ≤ 1 vanishes iff its residue does. □

Lemma 10 (γ is a difference across the collision pairing)**.** *For every* $\mathbf{R} \subseteq [N] \smallsetminus \{m\}$,

$$\gamma^{\nu}_{\mathbf{R}\cup\{m\}} = -\,\gamma^{\nu}_{\mathbf{R}}.$$

Moreover $\gamma^{\nu}_{\mathbf{M}} = 0$ *unless* $\mathbf{M} \le \nu$; *hence* $\gamma^{\nu}_{\mathbf{R}} = 0$ *unless* $\mathbf{R} \cup \{m\} \le \nu$.

Proof. The identity. Res_ν vanishes on $\mathcal{L}$: if $f \in \mathcal{L}$ then $\ell_\nu(f) \in \mathcal{O}$, so $t\ell_\nu(f) \in t\mathcal{O}$. By Proposition 8, $\mathcal{L} = W$, and the reduction map $W \to k^{\mathcal{P}}$ has image exactly $\mathcal{B} := \{g \in k^{\mathcal{P}} : g(\mathbf{M}) = g(\mathbf{M}^-)\ \forall \mathbf{M} \ni m\}$: it lands there by definition of W, and it is onto because given $g \in \mathcal{B}$ one may lift $g(\mathbf{R})$ arbitrarily for each $\mathbf{R} \not\ni m$ and then *define* $f(\mathbf{R} \cup \{m\}) := f(\mathbf{R})$, obtaining $f \in W$ with $\bar{f} = g$. Therefore

$$\sum_{\mathbf{M}} \gamma^\nu_{\mathbf{M}}\, g(\mathbf{M}) = 0 \qquad \text{for every } g \in \mathcal{B},$$

and $\mathcal{B}$ is free on the coordinates indexed by $\{\mathbf{R} : m \notin \mathbf{R}\}$. Taking g the element of $\mathcal{B}$ with $g(\mathbf{R}_0) = g(\mathbf{R}_0 \cup \{m\}) = 1$ and all other coordinates 0 gives $\gamma^\nu_{\mathbf{R}_0} + \gamma^\nu_{\mathbf{R}_0 \cup \{m\}} = 0$.

The support. By (M3), $\Phi_{\mathbf{MI}} = 0$ unless $\mathbf{I} \le \mathbf{M}$. Write $\Phi = D + L$ with $D = \mathrm{diag}(\widetilde{A}_{\mathbf{I},\mathbf{I}})$, so that $L_{\mathbf{MI}} \neq 0$ forces $\mathbf{I} < \mathbf{M}$ *strictly*. Then

$$\Phi^{-1} = \big(1 + D^{-1}L\big)^{-1} D^{-1} = \sum_{j \ge 0} (-D^{-1}L)^j D^{-1},$$

and the sum is finite because $\big((D^{-1}L)^j\big)_{\mathbf{IM}}$ is a sum over chains $\mathbf{M} = \mathbf{K}_0 < \mathbf{K}_1 < \cdots < \mathbf{K}_j = \mathbf{I}$ in the partial order (Corollary 6(1)), of which there are none for j exceeding the length of the longest chain. In particular that entry vanishes unless $\mathbf{M} \le \mathbf{I}$ *in the partial order*, so $(\Phi^{-1})_{\nu\mathbf{M}} = 0$ unless $\mathbf{M} \le \nu$. The last sentence then follows from the identity, since $\gamma^\nu_{\mathbf{R}} \neq 0$ forces $\gamma^\nu_{\mathbf{R}\cup\{m\}} \neq 0$ and hence $\mathbf{R} \cup \{m\} \le \nu$.

A step I had to strengthen. The one-line version of this — "Φ is triangular for a linear extension, and the inverse of a triangular matrix is triangular" — is **not enough**: it yields only $\mathbf{M} \preceq \nu$ for the chosen *linear* extension $\preceq$, which is weaker than $\mathbf{M} \le \nu$ and would not cut the support down to E^S. The chain expansion is what recovers the partial-order statement. I record the correction because the weaker form reads identically and the conclusion of the lemma is unchanged. □

Remark 11 (why this is the right object)**.** Lemma 10 is where the boundary family differs structurally from the pair family. There the collision multiplier is $g \neq 1$ and the analogous coefficients satisfy $\gamma_{\mathbf{M}} = -g\,\gamma_{\mathbf{M}''}$; here it is exactly 1, so Res_ν is a *plain* difference across the pairing $\mathbf{R} \leftrightarrow \mathbf{R} \cup \{m\}$. That is what lets the 2^N-term functional collapse to a sum over a combinatorially described set without any asymptotic input.

5 Motegi's inverse, and the one unproved input

Lemma 12 (Φ^{-1}, from all-cardinality biorthogonality)**.** *With* $|\lambda(\mathbf{A})| := \sum_{a \in \mathbf{A}} a$,

$$(\Phi^{-1})_{\nu\mathbf{M}} = q^{\,|\lambda(\nu)| - |\lambda(\mathbf{M})|}\, \frac{\widetilde{A}_{\mathbf{M}^{\mathrm{c}},\nu^{\mathrm{c}}}(w, q^{-1})}{\widetilde{A}_{\mathbf{M}^{\mathrm{c}},\mathbf{M}^{\mathrm{c}}}(w, q^{-1})\ \widetilde{A}_{\mathbf{M},\mathbf{M}}(w, q)}.$$

Proof. Motegi's `thm:BC-all-cardinality-biorthogonality` (L4365–4395) reads

$$\sum_{\mathbf{I} \subseteq [N]} q^{\,|\lambda(\mathbf{I})| - |\lambda(\mathbf{L})|} \frac{\widetilde{WX}_{N,N,|\mathbf{I}|}(w_{\mathbf{M}} | w|\mathbf{I})}{\Delta^X_{\mathbf{M}}(w,q)} \cdot \frac{\widetilde{WX}_{N,N,N-|\mathbf{I}|}(w_{\mathbf{L}^{\mathrm{c}}} | w|\mathbf{I}^{\mathrm{c}})\big|_{q \mapsto q^{-1}}}{\Delta^X_{\mathbf{L}^{\mathrm{c}}}(w,q^{-1})} = \delta_{\mathbf{M},\mathbf{L}},$$

i.e. $\mathsf{A}^X(\mathsf{A}^X)^\sharp = \mathrm{Id}$ where $\mathsf{A}^X_{\mathbf{M},\mathbf{I}} = \widetilde{WX}_{N,N,|\mathbf{I}|}(w_{\mathbf{M}}|w|\mathbf{I})/\Delta^X_{\mathbf{M}}$ and $(\mathsf{A}^X)^\sharp_{\mathbf{I},\mathbf{L}} = q^{|\lambda(\mathbf{I})|-|\lambda(\mathbf{L})|}\mathsf{A}^X_{\mathbf{L}^{\mathrm{c}},\mathbf{I}^{\mathrm{c}}}(w,q^{-1})$. Two identifications:

1. $\widetilde{WX}_{N,N,r}(w_{\mathbf{A}}|w|\mathbf{I}) = \widetilde{A}_{\mathbf{A},\mathbf{I}}$. Motegi pads $w_{\mathbf{A}}$ with $N - |\mathbf{A}|$ entries equal to 1 and takes $n = N$ (L4250–4253); by iterated stability (type C L2822–2828, type B L3602–3608) the padded value at $n = N$ equals the unpadded one at $n = |\mathbf{A}|$ when $|\mathbf{A}| \geq r$, and both vanish when $|\mathbf{A}| < r$ (L2843, L3620–3626).

2. $\Delta^X_{\mathbf{A}}(w, q) = \widetilde{A}_{\mathbf{A},\mathbf{A}}$. This is `eq:allcard-Delta-factor`, L4257–4262, *verbatim*: $\Delta^X_{\mathbf{A}} := \widetilde{WX}_{N,N,|\mathbf{A}|}(w_{\mathbf{A}}|w|\mathbf{A})$.

Hence $\mathsf{A}^X = \mathrm{diag}(\widetilde{A}_{\mathbf{M},\mathbf{M}})^{-1}\Phi$, so $\Phi^{-1} = (\mathsf{A}^X)^\sharp\,\mathrm{diag}(\widetilde{A}_{\mathbf{M},\mathbf{M}})^{-1}$, which is the display. □

Remark 13 (gate 0b of the brief, answered, including its arm 3)**.** The brief required me to verify this dependency first-hand before using it, and pre-registered **P2**: *does* $\widehat{\Phi}$ *of* `lem:allcard-Phihat-inverse` *specialise to my* $\widetilde{A}$*-restriction at type* A*?* **Arm 3 fired: $\widehat{\Phi}$ is a different object.** By `eq:allcard-Phihat-def` (L4287–4297) $\widehat{\Phi}$ is the *type-A*, fixed-cardinality matrix $\Phi_{\mathbf{K},\mathbf{I}}/\Phi_{\mathbf{K},\mathbf{K}}$, block diagonal in cardinality with unit diagonal — one of the two factors of $\mathsf{A}^X = \widehat{\mathrm{Br}}^X\widehat{\Phi}$. It is not a restriction of my type-B/C $\widetilde{A}$ and I did not use it. What *is* mine, up to the diagonal row scaling in Lemma 12, is the **parent** theorem's matrix A^X, and I verified the parent theorem first-hand against my own symmetrisation engine: **336/336** pairs $(\mathbf{M}, \mathbf{L})$, $N = 1, 2, 3$, both types, two random rational (q, w) points each (§9). That is a *different mechanism* from Motegi's proof (his is branching plus RTV stable-envelope uniqueness; mine is evaluating the symmetrisation formula), so it is corroboration rather than a second reading of one engine — and it discharges, from my side and at $N \leq 3$, Gap 1 of Rick's D^∞ document.

Hypothesis 14 (L^S: sharp triangularity after boundary restriction)**.** If $\mathbf{A} \geq \mathbf{B}$ and $m \notin \mathbf{A}$ then $\widetilde{A}_{\mathbf{A},\mathbf{B}}\big|_S \neq 0$ in k.

Hypothesis 14 is the boundary analogue of Rick's Lemma L (his §2bis), which gives the same sharpness without the restriction, by a degeneration $w_c = T^{-e_c}$, $T \to 0^+$, at $q = 0$. That degeneration cannot be run with w_m pinned at ε, so Hypothesis 14 is **not proved here**. It is measured: 699/699 instances at $N \leq 4$, both types, every m, every admissible ε, with the plant ($\mathbf{A} \not\geq \mathbf{B}$, where the value *must* vanish) firing 1179/1179 (§9). Every result below that uses it says so.

Lemma 15 (γ is nonzero on its combinatorial support)**.** *Assume Hypothesis 14 and* (BD)*. If* $\mathbf{M} \leq \nu$ *and* $m \in \mathbf{M}$ *then* $\gamma^\nu_{\mathbf{M}} \neq 0$*. Equivalently, by Lemma 10:* $\gamma^\nu_{\mathbf{R}} \neq 0$ *if and only if* $\mathbf{R} \cup \{m\} \leq \nu$*.*

Proof. Apply Lemma 12. By (BD), $\mathrm{ord}_S\,\widetilde{A}_{\mathbf{M},\mathbf{M}}(w, q) = 1$ since $m \in \mathbf{M}$, and $\mathrm{ord}_S\,\widetilde{A}_{\mathbf{M}^c,\mathbf{M}^c}(w, q^{-1}) = 0$ since $m \notin \mathbf{M}^c$ — note (BD) is stable under $q \mapsto q^{-1}$, which is an automorphism of $\mathbb{C}(q)$ fixing the notion of genericity. By Corollary 6(3), $\mathbf{M} \leq \nu$ gives $\nu^c \leq \mathbf{M}^c$, so by Hypothesis 14 applied with $\mathbf{A} = \mathbf{M}^c$ (which omits m) and $\mathbf{B} = \nu^c$, the numerator restricts to a nonzero element of k. Hence $\mathrm{ord}_S(\Phi^{-1})_{\nu\mathbf{M}} = -1$ exactly, and $\gamma^\nu_{\mathbf{M}} = \overline{t(\Phi^{-1})_{\nu\mathbf{M}}} \neq 0$. The "only if" half is Lemma 10. □

6 Theorem S: the residue of a boundary structure constant

Definition 16. For $\lambda, \mu, \nu \in \mathcal{P}$ and $m \in [N]$ put

$$E^S(\lambda, \mu, \nu; m) := \big\{\mathbf{R} \subseteq [N] \smallsetminus \{m\} \;:\; \mathbf{R} \cup \{m\} \leq \nu,\;\; \mathbf{R} \geq \lambda,\;\; \mathbf{R} \geq \mu\big\}.$$

Theorem 17 (the residue formula)**.** *Let χ be S-admissible, $\kappa = \chi(\varepsilon)$, and $f = \Psi_\chi \phi_\lambda \phi_\mu$. Then*

$$\mathrm{Res}_\nu(f) = (1-\kappa) \sum_{\mathbf{R} \in E^S(\lambda,\mu,\nu;m)} \gamma^\nu_{\mathbf{R}}\, \overline{\Psi_\chi(\mathbf{R})}\, \overline{\phi_\lambda(\mathbf{R})\phi_\mu(\mathbf{R})}.$$

Assuming Hypothesis 14, every summand is nonzero in k. Consequently:

1. (no pole) *if $\kappa = 1$ or $E^S(\lambda,\mu,\nu;m) = \varnothing$, then $\hat{c}^\nu_{\lambda\mu} \in \mathcal{O}$;*
2. (necessity, and simplicity) *if $\hat{c}^\nu_{\lambda\mu} \notin \mathcal{O}$ then $\kappa \neq 1$ and $E^S(\lambda,\mu,\nu;m) \neq \varnothing$, and the pole is simple;*
3. (sufficiency when the sum is a single term) *if $\kappa \neq 1$ and $|E^S(\lambda,\mu,\nu;m)| = 1$ then, assuming Hypothesis 14, $\hat{c}^\nu_{\lambda\mu}$ has a simple pole at S^ε_m.*

Items (1) and (2) do not use Hypothesis 14.

Proof. By Lemma 3, $\hat{c}^\nu_{\lambda\mu} = \ell_\nu(f)$, and $f \in \mathcal{O}^{\mathcal{P}}$ because Ψ_χ is a unit at S and $\phi_\lambda, \phi_\mu \in \mathcal{O}^{\mathcal{P}}$ by (M5). By Lemma 9, $\mathrm{Res}_\nu(f) = \sum_{\mathbf{M}} \gamma^\nu_{\mathbf{M}} \overline{f(\mathbf{M})}$. Group the 2^N terms into the 2^{N-1} pairs $\{\mathbf{R}, \mathbf{R}\cup\{m\}\}$ and apply Lemma 10:

$$\mathrm{Res}_\nu(f) = \sum_{\mathbf{R} \not\ni m} \gamma^\nu_{\mathbf{R}} \Big(\overline{f(\mathbf{R})} - \overline{f(\mathbf{R}\cup\{m\})} \Big).$$

Now compute $\overline{f(\mathbf{R}\cup\{m\})}$. By (BC), $\overline{\phi_\lambda(\mathbf{R}\cup\{m\})} = \overline{\phi_\lambda(\mathbf{R})}$ and likewise for μ; and $\overline{\Psi_\chi(\mathbf{R}\cup\{m\})} = \overline{\chi(w_m)} \cdot \overline{\Psi_\chi(\mathbf{R})} = \kappa\, \overline{\Psi_\chi(\mathbf{R})}$. So $\overline{f(\mathbf{R}\cup\{m\})} = \kappa\, \overline{f(\mathbf{R})}$ and

$$\mathrm{Res}_\nu(f) = (1-\kappa) \sum_{\mathbf{R} \not\ni m} \gamma^\nu_{\mathbf{R}}\, \overline{f(\mathbf{R})}.$$

Two vanishing conditions cut this down to E^S. First, by Lemma 10, $\gamma^\nu_{\mathbf{R}} = 0$ unless $\mathbf{R}\cup\{m\} \leq \nu$. Second, by (M3), $\phi_\lambda(\mathbf{R}) = 0$ identically unless $\mathbf{R} \geq \lambda$, and likewise for μ; so $\overline{f(\mathbf{R})} = 0$ unless $\mathbf{R} \geq \lambda$ and $\mathbf{R} \geq \mu$. The surviving indices are exactly $E^S(\lambda,\mu,\nu;m)$, which is the display.

Nonvanishing of the summands, under Hypothesis 14: $\gamma^\nu_{\mathbf{R}} \neq 0$ by Lemma 15; $\overline{\Psi_\chi(\mathbf{R})} \neq 0$ by admissibility; and $\overline{\phi_\lambda(\mathbf{R})} = \widetilde{A}_{\mathbf{R},\lambda}|_S \neq 0$ by Hypothesis 14 applied with $\mathbf{A} = \mathbf{R}$ (which omits m) and $\mathbf{B} = \lambda$, since $\mathbf{R} \geq \lambda$ — and likewise for μ.

(1) Either factor kills the expression; by Lemma 9 a vanishing residue means $\ell_\nu(f) \in \mathcal{O}$. (2) is the contrapositive of (1) together with $\mathrm{ord}_S\, \ell_\nu(f) \geq -1$ from Lemma 9. (3) A sum of one nonzero term is nonzero. □

Remark 18 (the single gap, named)**.** The only thing missing from a clean "$\iff$" at fixed χ is whether $|E^S| \geq 2$ nonzero terms can cancel. This is the exact analogue of the step Rick needed Theorem W and D^∞ for in the pair-divisor case, and I have not reproduced that machinery here. Two things keep the statement honest. First, §7 removes the gap entirely at the cost of a quantifier on χ, and the resulting statement is unconditional modulo Hypothesis 14. Second, I measured how often the question is even asked: $|E^S| \geq 2$ occurs in 2/16 firing triples at $N = 2$, 71/185 at $N = 3$, 1156/1886 at $N = 4$, 13890/18002 at $N = 5$ and 143692/165134 at $N = 6$, with $|E^S|$ as large as 32. So the census below is *not* a plant drawn from inside its own hypothesis class: it tests cancellation in a majority of its instances, and no cancellation occurred.

7 Theorem S′: the converse, with the quantifier where it belongs

The pole locus of $\hat{c}^{\nu}_{\lambda\mu}$ depends on χ — that is the content of `thm:C` of `2026-10-17-c2` — so a converse must either fix χ or quantify over it. Rick's pair-divisor theorem is about the untwisted constants, where no quantifier is needed; on the boundary family the untwisted constants have *no* poles ($\kappa = \chi(\varepsilon) = 1$ for $\chi = 1$), so the quantified form is the right statement, and it is provable.

Theorem 19 (the boundary converse)**.** *Assume Hypothesis 14 and* (BD)*. Fix N, X, m, $\varepsilon^h = 1$ and $\lambda, \mu, \nu \in \mathcal{P}$. Then*

$$E^S(\lambda,\mu,\nu;m) \neq \varnothing \iff \begin{array}{l}\text{there is an } S\text{-admissible } \chi \text{ with } \chi(\varepsilon) \neq 1 \text{ for which}\\ \hat{c}^{\nu}_{\lambda\mu} \text{ has a (simple) pole at } S^{\varepsilon}_m.\end{array}$$

Moreover, for each $d \geq N-1$ the set of χ in the family $\{\sum_{i=0}^{d} a_i v^i\}$ that fail *to produce the pole is contained in a proper Zariski-closed subset of the coefficient space.*

Proof. ($\Leftarrow$) is Theorem 17(2).

($\Rightarrow$) Work in $k = \mathbb{C}(q)(w_j : j \neq m)$ and let $\chi(v) = \sum_{i=0}^{d} a_i v^i$ with $a_0, \ldots, a_d$ indeterminates, $d \geq N-1$. Such a χ is S-admissible for a outside a proper closed subset (each $\chi(w_j)$ and $\chi(\varepsilon)$ is a nonzero linear form in a). By Theorem 17,

$$\mathrm{Res}_{\nu}(f) = (1-\kappa) \sum_{\mathbf{R}\in E^S} c_{\mathbf{R}} \prod_{r\in\mathbf{R}} L_r, \qquad L_r := \chi(w_r) = \sum_{i=0}^{d} a_i w_r^i, \quad c_{\mathbf{R}} := \gamma^{\nu}_{\mathbf{R}}\, \overline{\phi_\lambda(\mathbf{R})\phi_\mu(\mathbf{R})},$$

and each $c_{\mathbf{R}} \neq 0$ by Theorem 17. The $N-1$ linear forms $\{L_r\}_{r\neq m}$ are k-linearly independent in a: their coefficient matrix is the $(N-1)\times(d+1)$ Vandermonde block (w_r^i) with the w_r pairwise distinct and $d+1 \geq N-1$ columns, hence of full row rank. Extend $\{L_r\}_{r\neq m}$ to a basis of the space of linear forms in a; in the resulting polynomial coordinates the squarefree monomials $\prod_{r\in\mathbf{R}} L_r$, for distinct subsets $\mathbf{R} \subseteq [N] \smallsetminus \{m\}$, are distinct monomials, hence linearly independent over k. So $\sum_{\mathbf{R}\in E^S} c_{\mathbf{R}} \prod_{r\in\mathbf{R}} L_r$ is a *nonzero* polynomial in a whenever $E^S \neq \varnothing$. Its zero locus is a proper closed subset, as is $\{\kappa = 1\}$ (a hyperplane) and the non-admissibility locus; a point a outside their union gives an S-admissible χ with $\chi(\varepsilon) \neq 1$ and $\mathrm{Res}_{\nu}(f) \neq 0$, i.e. a pole, simple by Theorem 17(2). □

Corollary 20 (the pole-free renormalisations, per ν)**.** *Let χ be S-admissible for every m and every $\varepsilon^h = 1$. If $\chi(\varepsilon) = 1$ for all $\varepsilon^h = 1$ — condition* (SR) *of `2026-10-17-c2` — then no $\hat{c}^{\nu}_{\lambda\mu}$ has a boundary pole, for any N, any (λ,μ,ν). If $\chi(\varepsilon) \neq 1$ for some ε, then the set of ν for which a pole at S^{ε}_m can occur is the* ***up-set*** *generated by $\{\mathbf{R}\cup\{m\} : \mathbf{R} \geq \lambda, \mathbf{R} \geq \mu,\ m \notin \mathbf{R}\}$, and is empty exactly when no $\mathbf{R} \subseteq [N] \smallsetminus \{m\}$ dominates both λ and μ.*

Proof. The first half is Theorem 17(1). For the second, $E^S(\lambda,\mu,\nu;m) \neq \varnothing$ is by definition the condition that some $\mathbf{R}\cup\{m\}$ with $\mathbf{R} \geq \lambda, \mu$ satisfies $\mathbf{R}\cup\{m\} \leq \nu$, which is monotone in ν by Corollary 6(1). □

8 The A half: there is no converse, and why

Proposition 21. *Fix $i < j$ and let $H = \{w_i = w_j\}$ with local ring $\mathcal{O}_{ij}$. For* every *unit Ψ of the A-type GKM ring $\mathcal{G}_{ij}$ — in particular for $\Psi = \Psi_\chi$ with χ any ij-admissible renormalisation, and for $\Psi = 1$ — every structure constant of $\{\Psi\phi_{\mathbf{I}}\}$ lies in $\mathcal{O}_{ij}$. Hence there is no renormalisation in this class for which an A-type pole occurs, and the analogue of Definition 16 for the A-type family is the empty set for all (λ,μ,ν).*

Proof. This is Theorem B of `2026-10-18-atype-divisors-and-the-gkm-lattice.tex`, with `lem:unit` supplying $\Psi_\chi \in \mathcal{G}_{ij}^{\times}$; the mechanism is that $\mathrm{span}_{\mathcal{O}}\{\phi_{\mathbf{I}}\} = \mathcal{G}_{ij}$ is a *ring* (there the A-type collapse multiplier is $\rho^A = 1$, by the u-symmetry of Lemma A1 there, equivalently because $D(e_i - e_j) = \varnothing$ in the orbit classification of `2026-10-19-typeB` Lemma `lem:orbits`), and $\Psi\mathcal{G} = \mathcal{G}$ for a unit Ψ. A ring has regular structure constants. □

Remark 22 (this is not the same shape of statement as Theorem 19)**.** The asymmetry is structural and worth stating plainly. On $H_{e_i-e_j}$ the two colliding fixed points $\mathbf{M}$ and $\mathbf{M}\triangle\{i,j\}$ carry the *same multiset of w-values*, so Ψ_χ — indeed any element of $\mathcal{G}_{ij}$ — has collision multiplier 1 *for every* χ, and the lattice is a ring on the nose. On S_m^ε the value multisets differ by the single element w_m, and the multiplier of Ψ_χ is $\chi(\varepsilon)$ — which is 1 for the natural χ but not for all of them. The A family therefore has no free parameter left to break, while the S family has exactly one, and it is $\chi(\varepsilon)$. In the language of `lem:orbits`: $D(e_i - e_j) = \varnothing$ against $D(e_m) = \{m\}$. The grouping "S/A" in the work assignment collects a family with one degree of freedom together with a family that has none.

9 Verification

All computation is pure SymPy plus an exact truncated-Laurent-series linear algebra module over $\mathbb{Q}((t))$ written for this session (`proofs/code-1019c2-sa-converse/`). Orders of vanishing are computed by Gaussian elimination with minimal-valuation pivoting on the substitution $w_m = \varepsilon + t$, with q and the remaining w_j fixed rationals; the arithmetic is over $\mathbb{Q}$ with `Fraction`, so the only approximation is the truncation order, and the routines assert they never needed a coefficient beyond it. Every run **states its domain** and **gates on genericity** in the sense of Remark 4.

V1 — the structure-constant census (end-to-end, direct linear solve)

Domain: $X = C$ at $N = 2$ and $N = 3$ and $X = B$ at $N = 3$; every $m \in [N]$; every admissible ε (both signs in type C, $\varepsilon = 1$ in type B); twists $\chi \in \{1,\ \psi,\ v\psi(v),\ v,\ v+2\}$; and **all** 2^{3N} triples (λ, μ, ν), no sampling. $\hat{c}_{\lambda\mu}^{\nu}$ is obtained by solving $\Phi x = \Psi_\chi \phi_\lambda \phi_\mu$ — i.e. *without* using Lemma 12 or any part of §4.

run	arm	poles	TP	FP	FN
$N=2$, C, $m=1,2$, $\varepsilon=\pm1$	$\kappa=1$ (3 of 5 twists)	0	—	0	0
$N=2$, C, $m=1$, $\varepsilon=-1$	$\kappa=-1$ ($v\psi$, v)	11	11	0	0
$N=2$, C, $m=2$, $\varepsilon=-1$	$\kappa=-1$	5	5	0	0
$N=2$, C, $m=1$, $\varepsilon=+1$	$\kappa=3$ ($v+2$)	11	11	0	0
$N=2$, C, $m=2$, $\varepsilon=+1$	$\kappa=3$	5	5	0	0
$N=3$, C, $m=1$	$\kappa\neq1$	93	93	0	0
$N=3$, C, $m=2$	$\kappa\neq1$	62	62	0	0
$N=3$, C, $m=3$	$\kappa\neq1$	30	30	0	0
$N=3$, C and B, all m	$\kappa=1$	0	—	0	0

Every measured pole order was -1 (simple), as Theorem 17(2) requires; no order ≤ -2 occurred anywhere. The $\kappa = 1$ rows are the negative control: the same twist $v\psi(v)$ that produces 93 poles at $\varepsilon = -1$ produces 0 at $\varepsilon = +1$, and $v+2$ does the reverse — so the instrument is not simply reporting "pole" whenever $E^S \neq \varnothing$.

V2 — the structural lemmas

- **Corollary 6(3)**, exhaustive: 21844/21844 pairs $(\mathbf{M}, \nu)$, all $N \le 7$, 0 failures.
- **Hypothesis 14**: 699/699 instances with $\mathbf{A} \ge \mathbf{B}$, $m \notin \mathbf{A}$; domain $N \le 4$, $X \in \{B, C\}$, every m, every admissible ε, gated-generic point. **Plant**: the same quantity for $\mathbf{A} \not\ge \mathbf{B}$ must vanish by (M3) — 1179/1179. The plant is the positive control that the instrument can see a zero at all.
- **Lemma 12** against a numerically inverted Φ: 448/448 entries, $N = 2, 3$, $X = C$, every m, both ε, every ν.
- **Lemma 10** ($\gamma_{\mathbf{R}\cup\{m\}} + \gamma_{\mathbf{R}} = 0$): 224/224, same domain.
- **(BD)** in type C: $v_S(\widetilde{A}_{\mathbf{I},\mathbf{I}}) = [m \in \mathbf{I}]$ for every $\mathbf{I}$, by *two* mechanisms (the symmetrisation quotient and the independent positive-root product `diag_rootproduct`), $N = 2, 3$, all m, both ε: agreement on every $\mathbf{I}$. In type B: 34/34, $N \le 3$.

V3 — the gate: Motegi's all-cardinality biorthogonality

Domain: $N = 1, 2, 3$; $X \in \{B, C\}$; two random rational (q, w) points each; all 4^N pairs $(\mathbf{M}, \mathbf{L})$. Result: **336/336**, 0 mismatches. The check evaluates $\widetilde{A}$ from the symmetrisation formula, so it is independent of Motegi's branching-plus-RTV proof.

V4 — the non-cancellation step, isolated

Domain: $N = 3$ and $N = 4$; $X = C$; every m; both ε; all triples; four twists, including one in which $\chi(w_j)$ are *independently prescribed* rationals — the computational counterpart of the interpolation in Theorem 19. The residue sum of Theorem 17 is evaluated term by term in k and tested against $E^S \ne \varnothing$. $N = 3$: 185 firing triples per twist, 0 FP, 0 FN, for each of four twists — and 71 of the 185 have $|E^S| \ge 2$ (up to $|E^S| = 4$), so cancellation was genuinely tested and did not occur. $N = 4$, **complete**: 1886 firing triples per twist summed over m (736, 562, 393, 195 for $m = 1, 2, 3, 4$), five twist/ε arms with $\kappa \ne 1$ per m, hence **9430/9430** true positives, 0 FP, 0 FN — of which 1156 of the 1886 triples have $|E^S| \ge 2$, up to $|E^S| = 8$. **Provenance, stated:** V4 *uses* Lemma 12, so it is not an independent check of that lemma — it is an independent check of the *non-cancellation step*, conditional on Lemma 12, which V2 and V3 verify separately.

Two instrument faults, reported

1. **A non-generic sample read as a phenomenon.** The first $N = 3$ census reported $\mathrm{ord}_S \det \Phi = 8$ against the predicted 4. The random point had $w_3 = 7/3$ with $q = 3/7$, i.e. $w_3 = q^{-1}$, so the root factors $(1 - qw_m w_3)$ and $(1 - qw_3/w_m)$ vanish at $w_m = +1$; the extra order appeared for exactly the four $\mathbf{I}$ predicted by that diagnosis. The control caught it because the predicted number $\#\{\mathbf{I} \ni m\}$ had been written down *before* the run. Fixed by the genericity gate of Remark 4.
2. **A simplifier that guesses.** The first biorthogonality run reported 20 mismatches at $N = 3$, all containing factors like $2^{41/58}$. Cause: `sympy.nsimplify` applied to a rational number, which searches for a "nice" closed form and returns algebraic nonsense. `nsimplify` is not a simplifier. Replaced by `Rational(cancel(·))`: 336/336. Both faults produced

confident, legible, wrong numbers; one read as a catastrophe and one as a phenomenon, and the sign of the error said nothing about its cause.

10 Gaps, graded

G1 — Hypothesis 14 (L^S): OPEN, measured. $\mathbf{A} \geq \mathbf{B}$, $m \notin \mathbf{A} \Rightarrow \widetilde{A}_{\mathbf{A},\mathbf{B}}|_S \neq 0$. Needed for Lemma 15, for the "every summand is nonzero" clause of Theorem 17, and hence for Theorem 19; *not* needed for Theorem 17(1) or (2). 699/699 with a 1179/1179 plant. The natural proof is Rick's Lemma L degeneration $w_c = T^{-e_c}$ at $q = 0$; it does not obviously survive pinning $w_m = \varepsilon$, and I did not find a replacement. **This is the first thing to attempt next.**

G2 — non-cancellation at fixed χ: OPEN, measured, and bypassed by a quantifier. Whether $|E^S| \geq 2$ nonzero summands can cancel for a *given* χ. Theorem 19 removes the gap by quantifying over χ; at fixed χ the statement remains a one-way implication plus the $|E^S| = 1$ case. Measured: 0 cancellations in 185 opportunities at $N = 3$ (four twists) and 0 in 1886 at $N = 4$ (five twist/ε arms, 9430 instances), with $|E^S| \geq 2$ in 38% and 61% of them respectively.

G3 — (BD) in type B: measured, not proved. $\operatorname{ord}_S \widetilde{A}^B_{\mathbf{I},\mathbf{I}} = [m \in \mathbf{I}]$ on $\{w_m = 1\}$. Type C is proved inside `prop:short` of `2026-10-17-c2`; the type-B diagonal is `normalizationWB` (L1386) and the needed count is 34/34 at $N \leq 3$. Everything in §§4–7 is therefore **proved in type C and conditional on G3 in type B.**

G4 — the $\varepsilon = -1$ component in type B: does not exist, not unproved. $\{w_m = -1\}$ is not a root hyperplane of B_N and (BC) fails there (`2026-10-19-typeB` §"beyond", 0/9 at $r \geq 1$ against 6/6 at $r = 0$). Recorded so that the asymmetry between the two types is not later read as a missing case.

G5 — Motegi's biorthogonality is cited, and now corroborated, not re-derived. Lemma 12 rests on `thm:BC-all-cardinality-biorthogonality`. I have read the statement and its setup first-hand (L4146–4411) and the start of its proof (`lem:allcard-Phihat-inverse`, L4424–4440), and verified the statement numerically against my own engine at $N \leq 3$ (336/336). I have *not* read the RTV stable-envelope uniqueness argument it ultimately rests on. This is the same dependency that sits under Rick's D^∞ and his Theorem W det lemma; it is now the one shared, unverified root of three results and should be graded as such in both registries.

11 Pre-registration: which arm fired

The brief pre-registered three questions with an explicit third, *not-well-posed* arm, because on 2026-10-19 cycle 1 all three binary registers returned an answer that was not on the ballot. Reported in full, including arm 3.

P1 — "the S/A converse holds at full q for all N". SPLIT: arm 3 for the A half, arm 1 for the S half. The A family carries no converse because it carries no poles under any renormalisation (Proposition 21); the S family does, and it is Theorem 19. **The premise of the question — that "S/A" is one family with one answer — is what failed.** The

arm-3 detector the brief specified (check each candidate divisor is a root hyperplane) fired as predicted, but on the S half and in type B only: $\{w_m = -1\}$ is not a root hyperplane of B_N (G4).

P2 — "$\widehat{\Phi}$ specialises to my $\widetilde{A}$-restriction at type A". Arm 3: $\widehat{\Phi}$ is a different object — the type-A, fixed-cardinality, unit-diagonal factor of A^X (Remark 13). The gate's purpose was served instead by the parent theorem, whose matrix *is* mine up to a diagonal row scaling, because $\Delta^X_{\mathbf{A}} = \widetilde{A}_{\mathbf{A},\mathbf{A}}$ by definition. Verified 336/336.

P3 — "the colength 2^{N-2} falls out of biorthogonality in one step, canonically". Arm 3: "canonically" has no content here, exactly as the brief anticipated. The boundary colength does fall out in one step — $\operatorname{ord}_S \det \Phi = \sum_{\mathbf{I}} \operatorname{ord}_S \widetilde{A}_{\mathbf{I},\mathbf{I}} = \#\{\mathbf{I} \ni m\}$, inside the proof of Proposition 8 — but that is the same count by the same index set, so there is no second index set for the first to be canonically isomorphic to. What biorthogonality actually bought was not the colength but Lemma 15: the *non*-vanishing of γ on its support, which no colength argument gives.

Remark 23 (a prediction recorded for next session)**.** Theorem 17 makes E^S the *only* combinatorial input to the boundary residue. It is therefore falsifiable in a sharp way: if Hypothesis 14 fails for some $(\mathbf{A}, \mathbf{B}, m, \varepsilon)$ then the corresponding $\gamma^{\nu}_{\mathbf{R}}$ or $\overline{\phi_\lambda(\mathbf{R})}$ vanishes and a triple with $E^S \neq \varnothing$ will show *no* pole for *every* χ. So a single FN in V1 or V4 at fixed χ that survives a change of χ refutes L^S, while an FN that moves with χ is G2 (cancellation). **The two open gaps are distinguishable by one experiment**, and that experiment is the cheapest next step after attempting L^S directly.

Questions opened

None. The allocator `memory/questions/NEXT-FREE.md` was not advanced by this session: G1 and G2 are gaps in a result, not new questions, and Q530 was answered as part of gate 0b (Remark 13, P3) rather than re-opened.